

SISD-KOPM: A Publication Model for Computational Capabilities

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GOAL

Inform Cases with Scientific Knowledge at Scale

When it is actionable, scientific knowledge creates societal value by informing use cases¹¹ where it is relevant. Increasingly, that application can occur through computational capabilities that can be applied to inputs and generate outputs according to evidence-based computational logic. This work focuses on organizing computational capabilities so they can be applied at scale to systematically inform all relevant use cases with scientific knowledge.

STRATEGY

Publish Computational Capabilities

Achieving this overall scalability goal requires more than adding actionable scientific evidence to knowledgebases. It requires a publication strategy for formally organizing evidence-based computational capabilities so they can be individually identified, evaluated, governed, and managed throughout their lifecycles. Knowledgebases do not normally do this. They are not typically organized according to the *computational capabilities* they enable or support.

Publication holds some promise in this regard. While traditional scientific publications communicate new findings, published computational capabilities are fundamentally different because of requirements to version them over time, corroborate them through intrinsic and external evidence, computational exercises, and repeatedly applied them in operational settings to inform use cases.

When actionable scientific knowledge can be formally represented using computational logic to arrive at specified computational capabilities – and has sufficient importance, utility, and enduring value – the strategy here is to publish such evidence-based computational capabilities. These publications should take the form of persistent scholarly and operational computational objects around which evidence, identity, referenceability, corroboration, governance, and versioned evolutions all accumulate. Published computational capabilities should provide clear narrative expressions and directly deployable realizations of their computational logic.

PRECURSORS

The Original Knowledge Objects

Prior work on how to organize, manage, and share unique computable knowledge artifacts focused on modeling the items that must be explicitly represented, organized, and related to one another within units of artifact-bearing *Knowledge Objects* (KOs).

The primary items in KOs are narrative and executable representations of computable knowledge. The KO model adds formal service descriptions controlling how computable knowledge gets invoked.

Computable knowledge cannot be adequately understood through its representations alone. Therefore, software tests, documentation, and metadata with identifiers are also parts of the KO model, enabling KOs to be described and their application verified and tracked.

Applying computational capabilities to inform real-world use cases often depends on families of artifacts. The KO model supports item-level multiplicity inside a single KO, and an extension of the KO model supports Knowledge Assemblies that aggregate KOs into larger computable knowledge structures.

The architectural evolution of the KO model transformed it from one that originally promoted KOs as digital objects for storing computable knowledge artifacts into an increasingly rich model for mobilizing computable knowledge with the help of other content.

However, it may turn out that *computational capability* is a more effective organizing concept than the current concept of *computable knowledge*. Rather than focusing on WHAT KNOWLEDGE IS, computational capabilities focus on WHAT CAN BE COMPUTED WITH KNOWLEDGE EXPRESSED IN LOGIC. This is the difference between organizing around a capability to compute predictions versus organizing around a predictive model itself.

Reorienting the KO model to organize around computational capabilities first and foremost aims to make evidence-based computational capabilities directly discoverable, referenceable, reusable, and applicable at scale. In a world of AI agents, this is an attractive proposition.

¹¹ In this context, a “use case” includes any decision situation, analytic task or other similar event that while being experienced or performed can be informed by external scientific information.

NEW POTENTIALS

Publication Brings New Possibilities

Shifting from Packaging to Publishing

Packaging and publishing are fundamentally different activities. In the context of this work, packaging is an engineering activity concerned with constructing software artifacts for distribution and deployment. In contrast, publishing is a scholarly activity concerned with creating durable, identifiable, versioned, and corroborated objects for scholarly communication, evaluation, governance, deployment, operation, and reuse. Publication of computational capabilities must be done in a manner that supports their own reproducibility while communicating their identity, provenance, evidence, scientific meaning, deployment requirements, normative representations, and executable realizations.

Imagining a New Type of Publication

While first viewed as a compound digital object, the KO has at times been recast as a software repository, a metadata manifest, and a software package. These shifting perspectives now suggest another possibility. Rather than viewing the KO principally as a structured package of computable knowledge, it can instead be viewed as a formal stand-alone unit of publication for computational capabilities. Its purpose would then be to provide evidence-based computational capabilities in direct ways that support their real-world application at scale.

Envisioning Canonical Publications

Formal publication also has the potential to produce canonical published computational capabilities with equally canonical executable realizations. For such publications to become canonical, they must have unique identities. In addition, their provenance must always be clear, and they must be produced, corroborated, and governed in understandable ways.

For the executable realizations that come with published computational capabilities to become canonical, they must be realized through versioned releases. Although some technical adaptation will necessarily remain outside the publication boundary, canonical executable realizations should maximize direct application. Their value derives not only from faithfully realizing the published computational capability, but also from reducing unnecessary downstream reimplementations. Existing organizations and institutions have previously expressed some interest in providing and consuming canonical published computational capabilities.

CONTRIBUTION

Scope for a New Publication Model

Publication models specify the requirements that define classes of publications. This paper introduces the Specify-Implement-Serve-Deploy Knowledge Object publication model (SISD-KOPM). It models enhanced KOs with the intent of having published computational capabilities.

Like the IMRAD publication model for scientific investigations, the SISD-KOPM defines a unique publication type and its special requirements.

The SISD-KOPM goes beyond traditional publication models by intentionally combining the permanence and citability of scholarly publication with the versioned evolution of operational software. It establishes published computational capabilities as durable, structured, versioned and evolving scholarly objects that accumulate a variety of evidence related to their underlying scientific content and their real-world effects rather than existing only as software artifacts. As a complement to this paper, a normative architectural specification of the SISD-KOPM is defined in a separate document.

CENTRAL ORGANIZING CONCEPT

Computational Capability

For the purposes of the SISD-KOPM, an evidence-based computational capability is defined as *the ability to transform input use case data into output data relevant to the use case by applying computational logic based on scientific knowledge in a specified and reproducible manner*.

With computational capabilities as its central organizing concept, the SISD-KOPM aims to make published computational capabilities findable, accessible, interoperable and reusable (F.A.I.R.), thereby facilitating the large-scale application of evidence-based computational capabilities to inform relevant use cases with scientific knowledge.

DESIGN PRINCIPLES

Insights Drawn from Prototyping

Incremental Realization

In the SISD-KOPM, evidence-based computational capabilities are realized incrementally across four layers. These are a fundamental specification layer plus three more layers to add executability, invocability, and deployability to a published computational capability (See figure). Each layer plays its unique role exclusively while preserving the published computational capability in a manner that can be verified across all layers.

Referenceability

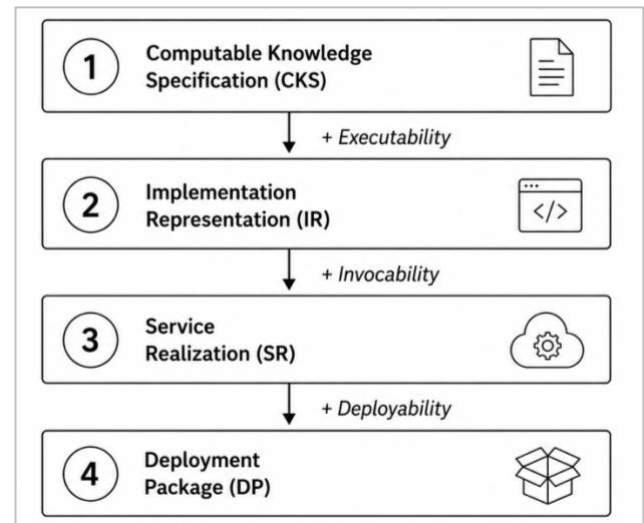
Referenceability aims to make every element of a published computational capability uniquely and persistently identifiable. Referenceable identities enable published computational capabilities, their layers, realizations, corroborating evidence, and other elements to be individually cited, discovered, inspected, governed, and related to one another throughout their lifecycles. By making every element referenceable in addition to the whole publication, the SISD-KOPM supports fine-grained scholarly communication, independent evaluation, version-aware reuse, and the accumulation of corroborating evidence around specific elements.

Corroboration

Corroboration establishes and continually strengthens confidence that realizations faithfully implement the published computational capability. The SISD-KOPM's full architectural specification supports a wide variety of interwoven corroboration elements that together make up a strong fabric of corroboration for any published computational capability. Intrinsic corroboration elements establish and maintain computational fidelity across layers through execution verification, equivalence analyses, conformance testing, and related evidence. Additional external corroborating evidence, such as independent evaluations, vulnerability analyses, performance assessments, and certifications can accumulate around a published computational capability and be explicitly associated with it.

Extended Operational Reach

In the SISD-KOPM, operational reach is defined as *the degree to which published computational capabilities participate directly across different operational environments*. With the advent of conversational and agentic AI, it has become possible to translate and transform one realization into other operationally



equivalent realizations that are suitable for deployment in a variety of common operational environments. As a principle, extended operational reach aims at enhancing the value of a published computational capability. By design, a published computational capability is a durable scholarly identity whose representational realizations may both accumulate and evolve without being limited to any fixed set of transient execution technologies.

Reproducibility from Specification

In the SISD-KOPM, every element that contributes to realizing a published computational capability gets specified in sufficient detail to permit its independent reproduction from the publication. All derivative executable elements can be computationally verified against the lower-level specifications from which they are derived, recursively through every realization layer down to the lowest one. Further, every realization is built up only from an immediately preceding layer of the realization hierarchy. Realizations that bypass intermediate layers or introduce alternative realization paths are not permitted. Realization path verification ensures that every realization remains strictly faithful to this critical design principle which exists to uphold reproducibility in scientific publishing.

Include Computational Exercises

In any published computational capability, computational exercises express how that capability is intended to be exercised across its realization hierarchy. Collectively, they should characterize the intended computational behavior of the published computational capability through representative, boundary, exceptional, and other scientifically meaningful use cases. These exercises provide common

points of comparison through which corroborating evidence can accumulate over time. Grounded in the published Computable Knowledge Specification, they exercise realizations derived from that specification and can help to corroborate their faithful implementation. For individual executable realizations, computational exercises can be run as software tests to confirm that external operational execution reliably applies the published computational capability. Publishing both the computational capability and its computational exercises strengthens the publication's ability to support independent verification, comparison, and long-term evolution.

A COHERENT UNIT OF PUBLICATION

Everything Necessary for Repeated Application

The four realization layers of the SISD-KOPM intentionally span the complete scientific and technical progression from specifying a computational capability as a narrative expression based on scientific knowledge and evidence through preparing it for its operational deployment.[¶] Together, these layers define a coherent publication unit that contains everything required to identify, understand, corroborate, realize, invoke, and prepare a computational capability for its operational application. Actual deployment, execution, and operational application intentionally remain outside the publication boundary, where they occur within local computational and institutional environments, which are governed by local policy and provide enterprise-approved computational capabilities.

CONCLUSION

A World of Preferred Computational Capabilities

The publication industry once transformed scientific communication and practice by creating durable units for communicating experiments and investigations. As applied computational capabilities increasingly become the means through which scientific evidence informs use cases, the next large-scale transformation of communication and practice is starting to emerge.

To support this transformation, a new publication model is warranted for better organizing, governing, corroborating, and repeatedly applying computational capabilities. Published computational capabilities represent a distinctly blended new class of scholarly

and operational publication whose many requirements extend beyond those of traditional scientific reports. The SISD-KOPM is proposed as one approach to meeting those many blended requirements.

If widely adopted, SISD-KOPM or similar new publication models could allow evidence-based computational capabilities to become shared scholarly infrastructure for computation itself. Rather than repeatedly redeveloping important computational capabilities from descriptive narrative specifications, developers, AI agents, and operational systems could increasingly discover, evaluate, reference, deploy, and apply canonical published computational capabilities in much the same way that today's software developers rely on widely trusted software libraries.

The ultimate success of a publication model for computational capabilities will be measured by the extent to which its published computational capabilities become the preferred capabilities that people, software, and AI agents repeatedly discover, reference, deploy, apply, and bolster.

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[¶] The "Ten Simple Rules" paper [2] provides practical guidance for making computable knowledge artifacts more findable, accessible, interoperable, shareable and reusable through thoughtful packaging, documentation, metadata, testing, and persistent identification. The SISD-KOPM builds on this foundation by defining a publication model whose purpose is to

publish evidence-based computational capabilities as durable scholarly and operational objects capable of accumulating evidence, identity, provenance, corroboration, and versioned evolution.